Ad Creator: AI-Powered Advertisement Image Generation

Project Report | Python command-line application

Abstract

Ad Creator is a Python command-line application for producing and refining advertisement visuals with text-to-image models. The project addresses the time and design effort required to prepare initial promotional artwork for different products and digital platforms. Its workflow collects product and brand details, combines them with category, visual-style and format presets, and builds a structured prompt with composition and quality guidance. Users can generate images through Stable Diffusion XL on Hugging Face or FLUX and SDXL Turbo through Pollinations.ai, then apply text, logo and image adjustments with Pillow. The application also provides a gallery and exports images as PNG, JPEG or WebP, including platform-size presets. Review of the implementation confirms a modular pipeline spanning configuration, prompt construction, model requests, post-processing and export. Seven generated PNG files were present in the supplied output folder, demonstrating that image generation had been exercised. A visual sample review found recognizable product imagery, alongside imperfect generated package lettering and a provider watermark in one banner sample. The project therefore provides a functional creative starting point, while accurate text, branding consistency and output quality still require human review.

1. Introduction

Digital advertising requires artwork that fits varied products, brand identities and platform dimensions. Creating each concept manually can be slow, especially when a team needs several visual directions before selecting a final design. This project explores how accessible text-to-image services can accelerate early-stage ad creation while retaining simple controls for editing and export. The topic was selected to combine generative AI with a practical, repeatable media workflow rather than image generation alone.

2. Problem Statement

Preparing promotional images often involves repeated prompt writing, model selection, resizing, branding and file conversion. A general image generator does not by itself organize these steps around advertising needs. Ad Creator addresses this workflow gap with reusable presets and a single guided CLI. Generated content can still contain inaccurate text or provider marks, so the tool supports finishing operations but does not replace creative review.

3. Objectives

- Generate product-focused advertising visuals from user-supplied details.
- Offer reusable style, category and platform-format choices.
- Support more than one model/provider for generation and comparison.
- Enable basic branding edits, image adjustments, gallery management and common-format export.

4. Methodology

Tools: Python; Hugging Face Inference API; Pollinations.ai; Pillow; Requests; python-dotenv. Generation uses external inference services, so network access is required; the Hugging Face option also requires a configured token.

Workflow: enter product details and choose a format, style, category and model; construct and preview an enriched prompt; request one or more images; optionally add copy or a logo and adjust the image; then save, browse or export the result. A seed can be supplied for repeatable requests where supported.

1. **Prompt Engineering:** A key challenge was generating images that look like advertisements rather than plain product photos. The prompt_builder module addresses this by combining: product details, ad-specific layout tokens (e.g., 'commercial advertisement poster design, graphic design layout with text space, marketing banner with headline area and call-to-action button'), visual style tokens, product category tokens, and negative prompts to avoid non-ad outputs. Aspect ratio detection adds appropriate composition hints.
2. **Image Generation Pipeline:** The generate_images() function handles the full pipeline: validating inputs, clamping dimensions to model limits (max 1,048,576 pixels, aligned to multiples of 8), routing to the correct provider based on model configuration, generating the requested number of images, and saving them with unique filenames. The compare_models() function generates with both SDXL and FLUX using the same prompt and seed for side-by-side comparison.
3. **Post-Processing:** The post_processor module provides: text overlay with headline, tagline, and CTA button on a semi-transparent band; logo placement at configurable corners with scaling; and image adjustments (brightness, contrast, saturation) via PIL ImageEnhance filters.

Module & Responsibility:

1. config.py - Model, style, category, ad-size and export presets.
2. prompt_builder.py - Combines product input with ad-layout, composition and quality guidance.

3. image_generator.py - Routes requests to model providers, constrains dimensions and saves generated images.
4. post_processor.py - Applies text overlays, logo placement, resizing and visual adjustments using Pillow.
5. exporter.py / main.py - Exports files and coordinates the interactive menus, gallery and workflow.

5. Results and Analysis

Evaluation: Both models were evaluated on their ability to generate advertisement-style images (not plain product photos).
Key findings:

- SDXL (Hugging Face): Produces high-quality, detailed images with strong composition. Best for photorealistic advertisements requiring fine detail and professional polish.
- FLUX (Pollinations): Excellent prompt adherence with a more artistic interpretation. Produces visually striking images with strong color and contrast. Completely free.
- Ad-specific prompt tokens significantly improved output quality, making images look like professional advertisements rather than plain product photos.
- The multi-model comparison feature allows users to evaluate which model produces better results for their specific use case.

Sample Result:

Test Case 1: Mango Juice - SDXL (Hugging Face)

Input Parameters:
- Product: Mango Juice (cold drink company)
- Keywords: New Launch, Real Mango Taste | Colors: Orange
- Format: Instagram Post (1080x1080) | Style: Emotional | Category: Food & Beverage
- Model: Stable Diffusion XL (Hugging Face) | Seed: 10

CLI Output:

```
Model: Stable Diffusion XL (Hugging Face)
Provider: Huggingface
Dimensions: 1024x1024
[1/1] Generating...
[API] Requesting image from Hugging Face...
[1/1] Saved: ad_20260928_171924_sdxl-hf_a766b8063b5b.png
Generated 1 image(s): 1024x1024
```

